

Self-Assessment Test Report — (Optimized Implementation)

Environment notice: these figures were collected on an environment compliant with guideline section 2.1 (Linux + GNU GCC + x86-64) and may be used as the official self-assessment result.

Auto-generated by `gen_report.py` from `reports/*.json` , following guideline section 3.5.

1. Algorithm Information

Item	Content
Category	Public-key cryptography
Function	Digital signature
Name	TSUOV
Implementation version	Optimized Implementation
Parameter sets / security levels	TSUOV_128 / TSUOV_256 / TSUOV_512 (claimed 128 / 256 / 512-bit security)
Programming interface	<code>sig_keygen</code> / <code>sig_sign</code> / <code>sig_verify</code>
Underlying hash	SM3 (mu derivation per parameter set)

2. Evaluation Environment

Item	This run (compliant)	Compliance requirement (section 2.1)
Processor	x86_64 (Linux)	64-bit x86, x86-ISA compatible, >2GHz recommended
Cores	single process / single thread	single core only
Memory	15 GB	>4GB recommended
Operating system	Linux 6.18.33.2-microsoft-standard-WSL2 [OK]	Linux, kernel >= 4.19
Compiler	gcc (Ubuntu 9.4.0-1ubuntu1~20.04.2) 9.4.0 [OK]	GNU GCC >= 8.5.0
Build tool	CMake 3.16.3 [OK]	CMake >= 3.11.4
Instruction set	x86-64 base (RDTSC timing), AVX2 available	x86-64 + AVX2
Start / end time	2026-06-29 15:45:59 ~ 15:46:21 (local)	fill in as measured

3. Functional Test

Method: for each security level, a key pair is generated from a fixed DRNG seed; a fixed 64-byte message is signed and verified; a tampered signature must be rejected. Full KAT vectors are in `API_PKC/Test_Vector/` and `data/kat/` .

Security level	keygen/sign/verify correctness	forged-signature rejection	Result
TSUOV_128	PASS	PASS	PASS
TSUOV_256	PASS	PASS	PASS
TSUOV_512	PASS	PASS	PASS

4. Performance Test

Each operation is run **500 times** (≥ 100); signing uses a fixed 64-byte message. The table reports average CPU cycles and throughput (ops/s).

Security level	Operation	Avg cycles	Throughput (ops/s)
TSUOV_128	KeyGen	5,206,568	560.52
TSUOV_128	Sign	6,243,822	467.40
TSUOV_128	Verify	5,505,839	530.05
TSUOV_256	KeyGen	32,983,645	88.48
TSUOV_256	Sign	38,956,734	74.91
TSUOV_256	Verify	36,580,096	79.78
TSUOV_512	KeyGen	438,368,587	6.66
TSUOV_512	Sign	361,843,558	8.07
TSUOV_512	Verify	412,087,359	7.08

5. Resource Consumption Test

Security level	Static text (B)	data (B)	bss (B)	Peak resident (B)
TSUOV_128	320	0	0	1,970,176
TSUOV_256	320	0	0	1,941,504

Security level	Static text (B)	data (B)	bss (B)	Peak resident (B)
TSUOV_512	320	0	0	2,383,872

Static segments are parsed from the algorithm static library (ELF sections); peak resident memory is measured via `getrusage(RUSAGE_SELF)` .

6. Transfer & Storage Overhead

Security level	Public key (B)	Private key (B)	Signature (B)
TSUOV_128	779	32	896
TSUOV_256	2,170	64	2,412
TSUOV_512	21,319	128	2,939

7. Raw Evidence Index

Item	Content
Test command	<code>run_self_assessment.sh 500</code>
Input data	fixed seed + 64B message (<code>data/README.md</code>); KAT vectors in <code>data/kat/</code>
Raw log	<code>logs/self_assessment.log</code>
Structured results	<code>reports/<algorithm>_<timestamp>.json</code> , <code>reports/summary.csv</code>

- Benchmark code: `bench/sig_bench.c` , adapter: `adapter/sig/` , entry: `src/main.c`
- Reproduction: run `run_self_assessment.sh` on a compliant host.